Experiment 1: Average Marks of Students Using NumPy

Aim

To calculate the average score of each subject and identify the subject with the highest average score using NumPy arrays.

Algorithm

1. Start the program.
2. Import the NumPy library.
3. Create a 4×4 NumPy array containing marks of four students in Math, Science, English, and History.
4. Calculate the average marks of each subject using np.mean(student_scores, axis=0).
5. Find the index of the highest average using np.argmax().
6. Display the average marks and the subject with the highest average.
7. Stop the program.

Code

```
import numpy as np

# Marks of 4 students in 4 subjects

student_scores = np.array([

   [85, 90, 78, 88],

   [92, 85, 80, 75],

   [78, 88, 92, 84],

   [88, 95, 85, 90]

])

subjects = ["Math", "Science", "English", "History"]

# Calculate average marks for each subject

averages = np.mean(student_scores, axis=0)

for subject, avg in zip(subjects, averages):

   print(f"{subject} Average: {avg:.2f}")

# Find the subject with the highest average

highest_index = np.argmax(averages)
```

```
print("Subject with the highest average:",
    subjects[highest_index])
print("Highest average score:",
    f"{averages[highest_index]:.2f}")
```

Input

Math	Science	English	History
85	90	78	88
92	85	80	75
78	88	92	84
88	95	85	90

Output

Math Average: 85.75

Science Average: 89.50

English Average: 83.75

History Average: 84.25

Subject with the highest average: Science

Highest average score: 89.50

```
========= RESTART: C:\Users\sivak\Desktop\FODS LAB experiments\Exp 1.py ========
Math Average: 85.75
Science Average: 89.50
English Average: 83.75
History Average: 84.25

Subject with the highest average: Science
Highest average score: 89.50
```

Result

Thus, the average marks of each subject were calculated using NumPy, and Science was identified as the subject with the highest average score of 89.50.